$$\sigma_x = \frac{\partial^2\Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2\Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2\Phi}{\partial x\,\partial y} - Xy - Yx$$

→ Φ está completamente equilibrado

modelo constitutivo (Modelo elástico lineal Hooke)

$$\varepsilon_{xx} = \frac{1}{E}\left(\frac{\partial^2\Phi}{\partial y^2} - \nu \cdot \frac{\partial^2\Phi}{\partial x^2}\right)$$

$$\varepsilon_x = \frac{\sigma_x - \nu\sigma_y}{E}, \quad \varepsilon_y = \frac{\sigma_y - \nu\sigma_x}{E}, \quad \gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\nu)\tau_{xy}}{E}$$

$$\frac{\partial^2\varepsilon_{xx}}{\partial y^2} = \frac{1}{E}\left(\frac{\partial^4\Phi}{\partial y^4} - \nu \cdot \frac{\partial^4\Phi}{\partial y^2\,\partial x^2}\right)$$

ecuaciones de compatibilidad

$\gamma_{xy} = 2\,\varepsilon_{xy}$

$$\frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2} = 2\,\frac{\partial^2 e_{xy}}{\partial x\,\partial y},$$

Φ función de tensión de Airy

Ecuación biarmónica

$$\nabla^4\Phi = \frac{\partial^4\Phi}{\partial y^4} + 2\cdot\frac{\partial^4\Phi}{\partial x^2 y^2} + \frac{\partial^4\Phi}{\partial x^4} = 0$$

$\Delta(\Delta\Phi) = \ldots$

DERIVANDO

$$\Phi(x,y) = ax^4 + bx^3y + cx^2y^2 + dxy^3 + ey^4$$

$4ey^3 \quad 12ey^2 \quad 24ey \quad 24e$

$$\frac{\partial^4\phi}{\partial y^4} = 0 + 0 + 0 + 0 + 24e$$

$$\frac{\partial^4\phi}{\partial x^4} = 24a + 0 + 0 + 0 + 0$$

$$\frac{\partial^4\phi}{\partial x^2 y^2} = 0 + 0 + \quad + 0 + 0$$

$2c\,2 = 4c$

$24e + 24a + 24c = 0$

$3e + 3a = -c$

$$\Phi(x,y) = ax^5 + bx^4y + cx^3y^2 + dx^2y^3 + exy^4 + fy^5$$

$\nabla^4\phi(x,y) = 0$

$\rightarrow (5a+b+c)\cdot x + (5\cdot f + b + d)\cdot y = 0$

si $x \neq 0$

$$c = \frac{-(5\cdot f + b + d)\cdot y - (5a+b)\cdot x}{x}$$

$$= -(5\cdot f + b + d)\cdot\frac{y}{x} - (5a+b)$$